Zhanhong He

Music and Audio Machine Learning | Generative Music Post-Training | Perceptual Evaluation

zhanh.he@research.uwa.edu.au | https://github.com/zhanh-he/ | https://www.linkedin.com/in/zhanhong-he/

Profile

Music and audio machine learning researcher with doctoral research in musical dynamics and research internships at Huawei and Dolby. Experience includes DPO/GRPO-based music-model post-training, music rewards and preference data, generative modelling, and human listening evaluation. Combines audio and symbolic music research with Python and PyTorch implementation.

Research And Industry Experience

Research Intern - Huawei Hongkong Research Centre | Started Jun 2026

- Applied DPO and GRPO in music-generation post-training, including rollout workflows and music-specific rewards.
- Developed rewards informed by music information retrieval and signal processing, and contributed to human-preference data collection and analysis.

Research Intern - Dolby Laboratories, Sydney | May-Sep 2024

- Developed music-generation models using U-Net, variational autoencoder and masked Transformer architectures.
- Designed and conducted professional human listening tests; related music-expression research is described in the CMMR 2025 project below.

Doctoral Research - University of Western Australia | Feb 2022-present

- Developed a multi-task CRNN for musical dynamics, beats, downbeats and structural-change estimation, and reimplemented a Bark-scale specific-loudness feature extractor in PyTorch.
- Investigated cross-instrument MIDI-velocity estimation using differentiable SoundFont proxies and audio-based supervision when velocity labels are limited.
- Developed a score-informed Transformer module to refine MIDI-velocity estimates from transcription models.
- Adapted a U-Net image-colourisation model to MIDI-velocity prediction and designed a listening evaluation with professional listeners.

Tutor and Lab Facilitator - University of Western Australia | Feb 2023-Dec 2025

- Taught Signal Processing and Engineering Practices, and co-supervised a Computer Science Honours project.

Selected Publication And Research

He, Z., Meng, H., Huang, D., and Togneri, R. “Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network.” ICASSP, 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

He, Z., Meng, H., Huang, D. D., and Togneri, R. “Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies.” arXiv:2608.08985, 2026. https://arxiv.org/abs/2608.08985

“Score-Informed Transformer for Refining MIDI Velocity in Automatic Music Transcription.” Research project. https://github.com/zhanh-he/score-informed-amt

“Filling MIDI Velocity using U-Net Image Colorizer.” Research project. https://github.com/zhanh-he/midi-velocity-colorizer

Education

PhD research in Electrical and Electronic Engineering - University of Western Australia | Feb 2022-present

Defense completed; graduation certificate expected November 2026. Formal award pending confirmation.

BSc (Honours), Engineering Science - University of Western Australia | Oct 2018-Dec 2021

First Class Honours | GPA 7/7 | WAM 87.7/100

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Technical And Musical Practice

Python; PyTorch; music information retrieval; audio and symbolic music processing; DPO/GRPO; music reward design; human-preference data; listening evaluation; U-Net, VAE and Transformer research implementations.

Ableton Live; Max/MSP; MuseScore. Guitar and piano as a music enthusiast.